CIP-A106 Laboratory 4

# Evidence Manifest and Findings Matrix

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## 1. Chronological evidence manifest

Columns: artefact, filename, SHA-256, source, time, analysis tool, and whether the item is in this package. Items marked "No" stay in the VM (live malware, raw memory and executable images are not submitted). Screenshots are listed as groups here and one per row in screenshots_manifest.csv. The same data is in evidence_manifest.csv.

| Time | Artefact | Filename | SHA-256 | Source | Tool | In package |
|---|---|---|---|---|---|---|
| 2026-10-07 22:53:06 | Clean snapshot | `Lab04_Clean_Baseline (VMware snapshot)` | `n/a (VM state)` | Snapshot Manager, before the archive entered the VM | VMware Workstation | Yes (screenshots E04, E05) |
| 2026-10-07 22:56* | VM configuration screenshots | `E01 to E03` | `see SHA256SUMS.txt` | Host screen capture of VM Settings | VMware Workstation | Yes |
| 2026-10-07 before 23:14 | Source archive (host) | `C:\iso_src\Sample 3.zip (host only), 779,839 B` | `6EE4E1834DE2CC07C9BB2DFD7FC90B64AF8ACA131A862C0D8676AF1349CA8606` | Course archive named in the brief, downloaded on the host | PowerShell Get-FileHash (host) | No (hash only) |
| 2026-10-07 before 23:14 | Transfer ISO (host) | `D:\sample3.iso` | `85B05D9BA00CD16DD370B628F8943D3C85EB87C23178CD30CCEECF00FB49974C` | Built on the host from the archive for one-way transfer | Host tools, PowerShell Get-FileHash | No (hash only) |
| 2026-10-07 23:14* to 23:17* | Transfer window screenshots | `E06, E07` | `see SHA256SUMS.txt` | CD/DVD drive connected to sample3.iso, then disconnected | VMware Workstation | Yes |
| 2026-10-07 about 23:14 to 23:20 | Source executable (O) | `original\sample.bin (1,217,024 B; SHA-1 70AD543FAECB496CA4C2318E0C8F81A8CBB8FB62; MD5 66F33597CBF097345C51891AB951B641)` | `0DA91175E7D72A7FF2BCB3FD93F2BA7BBE4045F9C4DEE5C9685C7FDF6DA622A6` | Extracted inside the VM from the archive (read-only original) | PowerShell Get-FileHash | No (live malware) |
| 2026-10-07 about 23:20 | Working copy (W) | `work\sample3.bin` | `0DA91175E7D72A7FF2BCB3FD93F2BA7BBE4045F9C4DEE5C9685C7FDF6DA622A6` | Copy of O, identical hash; used for every tool | PowerShell Get-FileHash, PEStudio, Ghidra, x32dbg | No (live malware) |
| 2026-10-07 23:35 to 23:38 | PEStudio screenshots | `E08 to E13` | `see SHA256SUMS.txt` | Host screen capture of the VM | PEStudio 9.61 | Yes |
| 2026-10-07 before 23:45:17 | Decompiled AutoIt script (D) | `exe2aut_out\script.au3 (514,737 B, 5,551 lines)` | `788D259BE4F29DE5865FA18C4B0089F6A1F0D10C910376250AE6924F1E5B6099` | Output of autoit-ripper 1.1.2 on W (substitute for Exe2Aut) | autoit-ripper 1.1.2 | No (obfuscated script retained in VM) |

| Time | Artefact | Filename | SHA-256 | Source | Tool | In package |
|---|---|---|---|---|---|---|
| 2026-10-07 23:45:17 to 23:45:26 | Decoded hex-literal strings | evidence\ 30_au3_decoded_strings.txt (excerpt in T4, T5) | not recorded at the time | Decoded 1,324 hex literals of script.au3 that are printable ASCII (76 of the first 80 lines pasted) | PowerShell 5.1 | Yes (excerpt, rebuilt text) |
| 2026-10-07 23:49:54 | Second snapshot | Lab04_PreDebug (VMware snapshot) | n/a (VM state) | Taken after static work, before any debugging | VMware Workstation | Yes (screenshot E14) |
| 2026-10-07 23:56:52 | Ghidra project record | Project lab04, program sample3.bin (SHA-256 recorded by Ghidra, 12.1.2) | matches 0DA91175...622A6 | Import of W, 263 functions after analysis | Ghidra 12.1.2 | Yes (screenshots E15, E16) |
| 2026-10-08 before 00:19 | Baseline file list | evidence\20_baseline_files.csv | 476D83A1B7835D9CA91ED156CEDE641BA3E1C74E627149A9F706C90EDF987DD6 | File baseline before the first run | PowerShell | No (retained in VM) |
| 2026-10-08 before 00:19 | Baseline process list | evidence\20_baseline_processes.txt | EE560871671E6361CFEA2ABC20CE1EA6D22E7CF6B87D8827C7144ADA60197C0A | Process baseline | PowerShell | No (retained in VM) |
| 2026-10-08 before 00:19 | Baseline Run keys | evidence\20_baseline_run_keys.txt | 92FDE6438637300E6BF7697941572765D573ACE69C03C83907AF6795A5D86E98 | HKCU and HKLM Run values | PowerShell | No (retained in VM) |
| 2026-10-08 before 00:19 | Baseline scheduled tasks | evidence\20_baseline_tasks.txt | A746B5CD178FF855162B7EE0936DFD3CF735CDC5FD9C9AA8E007E7F640A29D44 | Task names (not Disabled) | PowerShell | No (retained in VM) |
| 2026-10-08 00:12 to 00:13 | Ghidra screenshots (entry point, WININET imports) | E17, E18 | see SHA256SUMS.txt | Host screen capture | Ghidra 12.1.2 | Yes |
| 2026-10-08 00:17 to 00:27 | Debug run 1, PID 3844 (ScyllaHide as found) | E19 to E28 | see SHA256SUMS.txt | Host screen capture; exit code 0x0 at 00:27 | x32dbg (May 27 2026), ScyllaHide | Yes |
| 2026-10-08 00:45:07 | Post-run-1 time stamp | evidence\21_after_run1_time.txt | C0D62A7DE7B90D05814AE575136990A74E3394589646799244231E4B698DCE46 | Get-Date record | PowerShell | No (retained in VM) |
| 2026-10-08 00:45:09 | Post-run-1 new files | evidence\ 21_after_run1_new_files.txt (excerpt T6) | BDC3E8DDF892E45D2A2DB0DB38A9551F8828E23E2C77E04E5CC5FF9F392FB1F9 | Files written after 00:19 (only Recent .lnk, IE history container, WebCache) | PowerShell | Yes (excerpt, rebuilt text) |
| 2026-10-08 00:47:36 | Post-run-1 Run keys | evidence\21_after1_run_keys.txt | 92FDE6438637300E6BF7697941572765D573ACE69C03C83907AF6795A5D86E98 | Identical hash to the baseline file: no difference | PowerShell | No (retained in VM) |
| 2026-10-08 00:48:08 | Post-run-1 scheduled tasks | evidence\21_after1_tasks.txt | 88B2112200BA6A698CDCC70306E12EF111EA2FCB449E3484C40439D4071258DB | Text comparison unreliable (column wrapping), see report 6.4 | PowerShell | No (retained in VM) |

| Time | Artefact | Filename | SHA-256 | Source | Tool | In package |
|---|---|---|---|---|---|---|
| 2026-10-08 00:49 | Aborted start, PID 5576 | E29 | see SHA256SUMS.txt | ScyllaHide changed after launch; start killed, not used as evidence | x32dbg, ScyllaHide | Yes |
| 2026-10-08 about 00:5x | x32dbg breakpoint database | evidence\22_run1_x32dbg_db.dd32 | 674CEDF5EC162581252725FE9C4CBFD1EF71825EE0129B446F41C707D3EF4FF8 | Saved breakpoints after run 1 | x32dbg | No (retained in VM) |
| 2026-10-08 00:56 to 01:08 | Debug run 2, PID 3848 (ScyllaHide unticked) | E30 to E37 | see SHA256SUMS.txt | Host screen capture; dialog at 01:06; exit code 0x0 | x32dbg | Yes |
| 2026-10-08 after run 2 | Post-run-2 new files | evidence\23_after_run2_new_files.txt | 952A9BAD1CFFC081D90DB925D60AD8E0933A4727DB87BB5048EEDCBC70B56B25 | Files written after run 2 | PowerShell | No (retained in VM) |
| 2026-10-08 after run 2 | Post-run-2 task files | evidence\23_after_run2_task_files.txt | D551A91A2B99A0FE0A7A73D3BEF5BC348E49F7339A63764520755D722E4AB365 | System32\Tasks files written (Windows-owned tasks, after sample exit) | PowerShell | No (retained in VM) |
| 2026-10-08 03:47 to 03:48 | Run 2 end-state screenshots | E38 to E40 | see SHA256SUMS.txt | Host screen capture | x32dbg | Yes |
| 2026-10-08 04:51 to 05:09 | Ghidra reference screenshots | E41 to E46 | see SHA256SUMS.txt | Host screen capture | Ghidra 12.1.2 | Yes |
| 2026-10-08 05:12 to 05:16 | Debug run 3, PID 3784 (ScyllaHide unticked) | E47 to E52 | see SHA256SUMS.txt | Process created 05:12:12; parent PID 3092 (x32dbg.exe) | x32dbg | Yes |
| 2026-10-08 05:21:30 | Memory artefact (M) | memory\run3_blocked.dmp (97,956,974 B) | 3DFEF5111364B61EC8345DBC835964E1A88216D335B2114C7EBA026381EAD5E5 | Full dump of PID 3784 while the AutoIt dialog was shown | System Informer 3.2.25113.1114 | No (raw memory dump) |
| 2026-10-08 about 05:22 | Dump validation record | evidence\24_dump_validation.txt | D04E10B488B7F334ECB2C20C2EACE70E1202876FF5E1F5BC2ED26195014BD421 | PowerShell validation of M (MDMP header, five string searches, DOS header search; text in T9) | PowerShell 5.1 | Yes (rebuilt text, T9) |
| 2026-10-08 05:25:34 | Post-run-3 new files | evidence\26_after_run3_new_files.txt (T2) | 47F745757DB87941F213336476584175817 31F69E364483FA612DA77B07B4FB2 | Files written after 05:00 (no scripts, executables or shortcuts) | PowerShell | Yes (rebuilt text) |
| 2026-10-08 05:25:52 | Indicator counts | evidence\25_indicator_counts.txt (T1) | 772088FB4D38AD26805F60784635622F58C92A0248779B7D87BD4BCCB67021D7 | Case-insensitive counts of 18 patterns in script.au3 and the decoded strings | PowerShell 5.1 | Yes (rebuilt text) |
| 2026-10-08 05:27:39 | Script context record | evidence\27_script_context.txt (T3) | 319EEB44744EF92F92A38485B98917D7EFA2E2435A8AC647C26F572110A7A702 | Lines of script.au3 that contain Startup, CallWindowProc or GdiplusStartup | PowerShell 5.1 | Yes (rebuilt text) |
| 2026-10-08 05:27 | System Informer and PowerShell screenshot | E53 | see SHA256SUMS.txt | Host screen capture | System Informer, PowerShell | Yes |

# 2. Findings matrix

Every behavioural conclusion, IOC and ATT&CK mapping with its source artefact, confidence level and limitation. Confidence: High = observed directly by a tool; Medium = observed by one source only or partly decoded; Low = inference or unconfirmed.

| ID | Type | Finding / indicator | Source artefact | Confidence | Limitation |
|---|---|---|---|---|---|
| F01 | Identification | The executable is an AutoIt-compiled program (AutoIt runtime plus an encrypted SCRIPT resource) | PEStudio E10, E12; autoit-ripper output | High | AutoIt version only given as the range 2.XX-3.XX |
| F02 | Identification | Claimed Lokibot identity is neither supported nor excluded: no Lokibot-typical strings in the script or decoded strings and no payload observed | T1 (18 patterns, all 0 except Startup 3 and CallWindowProc 2); Ghidra; dump | Low (for identity) | Payload and configuration were never observed; script encrypted/obfuscated; execution blocked by the debugger check |
| F03 | Unpacking | autoit-ripper 1.1.2 recovered script.au3 (514,737 B, 5,551 lines) | exe2aut_out\script.au3, SHA-256 788D259BE4F29DE5… | High | Exe2Aut itself not used; processing time not recorded |
| F04 | Unpacking | The recovered script is heavily obfuscated (1,324 hex literals, junk code, variable-held API names) | T4, T3 | High | Counted by pattern, not by full review |
| F05 | Capability | Script embeds x86 machine code as hex strings with API names built on the stack (NtCreateMutant, NtOpenProcessToken, NtAdjustPrivilegesToken, NtProtectVirtualMemory, NtDelayExecution, CryptDecrypt, CoCreateInstance, LoadLibraryA, GetProcAddress, \KnownDlls32) | T4, T5 | Medium | First 80 decoded lines only; fragments reconstructed; no disassembly or execution |
| F06 | Capability | Script holds an [InternetShortcut] template and a WScript relaunch template with a random 5 to 15 second delay | T4 | Medium (text exists) | Not executed; no .url or .vbs file written in any run (T2, T6) |
| F07 | Capability | Persistence is only a candidate: function LEQMOPUPON takes $STARTUP; Startup appears 3 times (437, 533 and GdiplusStartup at 3186) | T3, T1 | Low | Static text; one of three hits is GdiplusStartup |
| F08 | Anti-analysis | FUN_00403b3a calls IsDebuggerPresent at 00403b7a; TEST EAX,EAX and JNZ to 0043d272 lead to MessageBoxA with "This is a third-party compiled AutoIt script." and the exit path | E46, E45, E43 | High | Static reading plus the run 2 and run 3 observations; EAX after the call not captured |
| F09 | Anti-analysis | With a debugger attached and ScyllaHide off, execution stopped at the message box and the process ended with exit code 0 after OK (run 2); run 3 was held at the dialog | E35, E37, E38 | High | Run 3 dialog not photographed; the analyst saw it |
| F10 | Anti-analysis | The other four IsDebuggerPresent call sites were located but not individually reviewed; 0043b2f4 is ATL library code | E42, E44 | Medium | Three sites unreviewed |

| ID | Type | Finding / indicator | Source artefact | Confidence | Limitation |
|---|---|---|---|---|---|
| F11 | Runtime | Run 1 (ScyllaHide as found) continued past the check: threads, two VirtualAlloc breakpoint hits, private copies of five system DLLs, NtQueryInformationProcess and NtSetInformationThread through the copies, exit code 0 | E28, E26, E25 | High (observations) | ScyllaHide profile was as found, not confirmed as instructor-approved; the dialog was not captured in run 1 |
| F12 | Runtime | Private copies of ntdll, kernel32, user32, advapi32, ole32 were loaded from SysWOW64 at new bases; this fits the \KnownDlls32 and Nt* name fragments in the script | E28, T5 | Low (interpretation) | Reason not established; call arguments not captured |
| F13 | Memory | VirtualAlloc (kernel32 76D765F0): run 1 logged 2 hits with no size or protection shown; runs 2 and 3 set the breakpoint and had 0 hits | E28, E34, E51 | High (counts) | Requested size and protection not recorded |
| F14 | Memory | No hardware breakpoint set: no allocated region was reached in the blocked runs | E51 | High | Run 1 did not stop at VirtualAlloc with arguments; method documented as not relevant |
| F15 | Memory | run3_blocked.dmp (97,956,974 B, SHA-256 3DFEF5111364B61E…) acquired from PID 3784, parent 3092 (x32dbg.exe), at 05:21:30 | T7; T9 (24_dump_validation.txt) | High (acquisition) | Captures the runtime at the dialog; no payload; raw dump not submitted |
| F16 | Host effects | No script, shortcut or executable was written outside Recent .lnk files and browser caches; Run keys unchanged after run 1 | T2, T6, 21_after1_run_keys | Medium | Run keys re-checked after run 1 only; task comparison inconclusive (report 6.4) |
| F17 | Network | No domain or IP address of a server found; WININET and WSOCK32 imports are runtime imports | T1, E18, E11 | Low | No packet capture; adapter Host-only and disconnected |
| I01 | IOC: File hash of sample (O and W) | 0DA91175…622A6 (SHA-256), MD5 66F33597…, SHA-1 70AD543F… | PowerShell, PEStudio, Ghidra (E08, E15) | High | High: identifies the file, not the family |
| I02 | IOC: Import hash (MD5) | 733BD1E4C7463E0FD92D0EA34CBFA027 | PEStudio indicators (E12) | High | High as a value; Low as family evidence (any file built from the same AutoIt runtime shares it) |
| I03 | IOC: Packaging marker | AutoIt compiled script (2.XX-3.XX); RCDATA SCRIPT 359,376 B, entropy 7.999 | PEStudio, autoit-ripper (E10, E09) | High | High |
| I04 | IOC: PE timestamps | Compiler stamp 2020-02-26 11:25:12 UTC; debug stamp 2015-07-10 09:42:07 UTC | PEStudio (E08) | Medium | Medium: stamps differ by almost five years, which fits a stock AutoIt runtime, not a bespoke build |
| I05 | IOC: String (dialog text) | "This is a third-party compiled AutoIt script." | Ghidra, x32dbg (E41, E37) | High | High: shown when a debugger is reported |
| I06 | IOC: API, anti-debug | IsDebuggerPresent call at 00403b7a (and four other call sites) | Ghidra, x32dbg (E42, E45, E35) | High | High for the check at 00403b7a; Medium for the others (not reviewed) |

| ID | Type | Finding / indicator | Source artefact | Confidence | Limitation |
|---|---|---|---|---|---|
| I07 | IOC: Imports, network capability | HttpOpenRequestW, HttpQueryInfoW, HttpSendRequestW (WININET); socket, connect, send, recv (WSOCK32) | Ghidra, PEStudio (E18, E11) | Low | Low as an indicator: present in every compiled AutoIt executable that links the runtime; no traffic observed |
| I08 | IOC: String pattern (IP) | 255.255.255.255 | PEStudio indicators (E12) | Low | Low: broadcast address, a literal in the AutoIt runtime; not a contact address |
| I09 | IOC: Network IOC | Domains, URLs, IP addresses of a command server | script.au3, decoded strings, Ghidra, dump (T1, T4) | Not found | Not found (0 hits for .php, Mozilla, User-Agent, POST, wininet, InternetOpen, HttpSendRequest) |
| I10 | IOC: Script function (line 437) | LEQMOPUPON ( $FILE , $STARTUP , $RES , $RUN ) | script.au3 (E53, T3) | Low | Low: name and parameters only; behaviour not reviewed or executed |
| I11 | IOC: Script string templates (decoded hex) | "[InternetShortcut]" with URL=file:/// and a VBS template (WScript.Shell, WScript.Sleep random 5000-15000, WshShell.Run) | decoded strings (T4) | Medium | Medium that the templates exist; Low that they run: no .url or .vbs file was written in any run |
| I12 | IOC: Script strings (lines 945, 1696, 3186) | CallWindowProc, CallWindowProcW, GdiplusStartup (variables holding API names) | script.au3 (E53, T3) | Medium | Medium: AutoIt pattern for running machine code; payload not run |
| I13 | IOC: Embedded code, API names | Embedded x86 code as hex literals (546, 530 and many 300-character strings) that spell API names on the stack: NtCreateMutant, NtOpenProcessToken, NtAdjustPrivilegesToken, NtProtectVirtualMemory, NtDelayExecution, CryptDecrypt, CryptDestroyHash, CoCreateInstance, LoadLibraryA, GetProcAddress, \KnownDlls32\kernel32.dll, user32.dll | decoded strings (T4, T5) | Medium | Medium that machine code is embedded; Low for purpose (names are partial; code was not disassembled) |
| I14 | IOC: Runtime module addresses | Private copies of advapi32, ntdll, kernel32, user32, ole32 at 050B0000, 05140000, 05310000, 05410000, 055E0000 | x32dbg log (run 1) (E28) | High | High (observed); Low for the reason |
| I15 | IOC: Observation point and hits | kernel32.VirtualAlloc 76D765F0 (two breakpoint hits in run 1) | x32dbg (E28, E34, E51) | High | High that the hits happened; arguments not captured |
| I16 | IOC: Host IOCs | Mutexes, registry paths, dropped file names | baselines, post-run checks (T2, T6, 21_after1_run_keys) | Not observed | Not observed (no new scripts, shortcuts or executables; Run keys unchanged after run 1) |
| A-T1027 | ATT&CK T1027 | Obfuscated Files or Information: Encrypted SCRIPT resource, hex-encoded strings and code, junk code, API names held in variables | script.au3, T1, T4, E10 | High | Obfuscation is observed; its purpose in this sample is inferred |
| A-T1622 | ATT&CK T1622 | Debugger Evasion: IsDebuggerPresent then MessageBoxA and exit when a debugger is attached; execution stopped at the dialog in runs 2 and 3 | Ghidra FUN_00403b3a, E35, E37 | Medium | The check is part of the stock AutoIt runtime and not shown to be added by the author |

| ID | Type | Finding / indicator | Source artefact | Confidence | Limitation |
|---|---|---|---|---|---|
| A-T1106 | ATT&CK T1106 | Native API: Calls to NtQueryInformationProcess and NtSetInformationThread observed in run 1 through private copies of ntdll; Nt* names spelled in embedded code | E28, E26 | Medium | Arguments not captured; one run, ScyllaHide on |
| A-T1055 | ATT&CK T1055 (candidate) | Process Injection: Nt* names for process and memory handling and CallWindowProc in a script that embeds machine code | T3, T5 | Low | No injection observed; candidate only |
| A-T1547 | ATT&CK T1547 / T1112 (candidate) | Persistence by Run key or registry: Function LEQMOPUPON with $STARTUP, NtSetValueKey and RtlFormatCurrentUserKeyPath fragments | T3, T5, 21_after1_run_keys | Low | Run keys unchanged after run 1; runs 2 and 3 were stopped by the dialog; not mapped as supported |